

Pablo Nuñez Martinez

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First-year M.S. Aeronautics and Astronautics student with **hands-on and leadership** experience in projects involving **electric vehicles and aircraft**. Founder of a university electric racing team, specializing in **aerodynamics, FEA and CFD simulations, and advanced composite materials**. Passionate about sustainable mobility, motorsports and aerospace engineering, with strong skills in product design, manufacturing, and collaborative project management.

Education

Stanford University <i>Master of Science, Aeronautics and Astronautics. GPA 4.1/4.3</i>	Stanford, CA <i>Graduation Date - June 2027</i>
Tec de Monterrey <i>Bachelor of Science, Mechanical Engineering. GPA 97.19/100</i>	Guadalajara, Jalisco <i>June 2025</i>
McGill University <i>Study abroad, Aerospace Engineering. GPA 4.0/4.0</i>	Montréal, QC <i>August 2024 – December 2024</i>

Experience

Stanford Solar Car Project - World Solar Challenge <i>Aerodynamics Lead Engineer</i>	Stanford, CA <i>September 2025 – Present</i>
- Developed a parametric CAD design for the aerobody of the car to improve the efficiency of our iteration process and make sure that our design meets all the competition regulations. - Iterated aerodynamic designs using ANSYS Fluent and Luminary Cloud, analyzing flow separation, pressure distribution, wall shear stress, and aerodynamic performance, with a focus on minimum drag and crosswind stability. - Correlated wind tunnel data from previous years to define and validate CFD simulation parameters, including turbulence, transition models, and ground modeling, achieving >95% correlation between CFD and experimental results. - Led a 10+ member aerodynamics subteam, managing tasks and timelines in Notion, training new members in aerodynamics and CFD fundamentals, and maintaining comprehensive documentation of research and design workflows.	
ELYOS Racing Team - Shell Eco-marathon <i>Founder and Team Leader</i>	Guadalajara, Jalisco <i>December 2021 – June 2025</i>
- Designed and performed FEA simulations for composites, achieving a final car weight of 30 kg through different layups in each section of the car, implementing honeycomb as our core material for the laminate and balsa inserts in stress-concentrating sections. - Designed and prototyped three different steering systems using topology optimization to reduce weight and manufactured them through CNC machining and 3D printing with Nylon 6 + CF filament. - Conducted CFD analysis in Ansys Fluent to improve the drag coefficient of the car from 0.20 to 0.06 by optimizing the shape of the car using the NACA-00 airfoil family in specific parts of the car. - Manufactured two carbon fiber monocoques: first via wet layup, later via resin infusion with vacuum. - Led a multidisciplinary team of 50+ students across aerodynamics, structures, electrical, and business divisions; managed timelines, resources, and technical reviews throughout four competition cycles.	
McGill Formula Electric – Formula SAE Electric <i>Structures Engineer - Chassis</i>	Montréal, QC <i>August 2024 – December 2024</i>
- Worked on a design proposal for a monocoque chassis insert using balsa wood and G10 combo for weight reduction and an increase in compressive strength using Siemens NX. - Simulated and designed a downforce-generating device on the underbody of the car using Star CCM. - Learned about laminates theory and generated an optimization code using genetic algorithms to find the best angle combinations in certain load scenarios.	

Skills & Interests

Technical: Ansys Fluent, Star-CCM+, Luminary, Ansys Mechanical, LS-DYNA, Altair, SolidWorks, NX, Fusion 360, MATLAB, Python, Julia
Language: English (TOEFL C1), French (DELFB2), Spanish (Native).
Certifications: Mechanical design Professional (SolidWorks), Simulation (SolidWorks), Flight mechanics (ISAE-SUPERAERO).